

Different Phases of a Product Cycle

1. Ideation phase
2. Feasibility study
3. Design and development
4. Testing and verification
5. Collateral production
6. Manufacturing
7. Feedback and improvement

make a quick buck in a big market like India, or do a little work and export the product to other big markets and make some more money. On the other hand, there are companies who know that a product is merely not a product but a cycle that has the ability to keep end-consumers hooked not only to using the product but also wait in anticipation for the new variants of the same product to hit the market.

A typical product cycle, as per Sanket Bandyopadhyay, co-founder & VP, Daikoku Innovations, includes many phases (see box). He says the real improvement in the product comes after the manufacturing phase, and hence a company outsourcing the design and manufacture, or the design and manufacturing agency, should not consider the manufacturing phase as the final phase.

If you are looking to outsource design to a design house, you should provide them with all the specs you desire in the product right in the beginning. Adding specs down the line can hamper the advancements made in the design cycle. For example, if you wanted to source design of a compact looking electric vehicle charger that has the ability to fast charge also, you must specify whether your plans include deploying the same at homes or deploying the same on a commercial level. Imagine how big a mistake would it be if you got a charger designed for personal use while you wanted it for commercial use, or vice versa!

“There are some key stages in product development. Most of the challenges come in the ideation phase. The customer usually keeps adding new specs in the later stages, which makes things a lot difficult,” says Sanket.

The ideation phase matters a lot

because this is where the design partners try to gauge the design challenges they would be facing to meet the set of certifications required for selling the product in different geographies. In case the company getting a product designed (new companies, generally) is uncertain about the certifications they would need, the design house should at least be provided with exact specs.

“The second most difficult phase after ideation is feasibility as this is where the existing patent infringement can take place. Then, in terms of importance, comes the testing phase. It is important because we have to make sure that all the specs submitted by a company for a product work perfectly,” added Sanket.

Constant feedback on the developments made by the design partner can help in cutting the amount of overall time required to take the product to its final stages. This is where both the company that has outsourced the requirement and the company working on the requirement have to play an equally important role. The former should keep checking about the developments while the latter should update the former about even the smallest of changes, issues, or implementations it has done.

“There is a saying in our industry, and it goes like this: Repeat orders make for perfect feedback. But to get those repeat orders we have to make sure that every phase in the product development is followed religiously. It is usually us, the design houses and the manufacturing partners, that have to remind customers about giving us feedback. It is as much your product as it is ours, and we need feedback to make it better for you,” says Mohandas Ukkandth, MD, Sparr Electronics.

A common example of the mismatch between the customer and design or manufacturing house can be found in wireless earbuds. Many of these hot-selling categories of smart wearables promise hours of battery life, but in actual conditions a lot of these fail to match what they promise.

Then comes the part where the two have to decide on the look and feel of the product. No matter how great a technical product you have

developed, if it does not look and feel good, there are chances people will prefer your competitor over you. In the case of electronics, this look and feel is gauged by the quality of enclosure material used and how well the same has been saturated with the design and functionality of the product.

A great example of the same can be seen in the smartphones you use. Will you prefer to click images using a 100-megapixel camera embedded in a Chinese smartphone or a 12-megapixel camera embedded in an iPhone?

“The amount of time spent on finalising specs and other details, including the requirement of certifications, plays an important role in making any product successful. While outsourcing electronics design and manufacturing one should always remember that, while designing electronics is a simple process, the additions like plastic enclosures need a lot of focus and energy,” adds Mohandas.

Sanket adds, “The more closely knit the customer and the manufacturer are with the design services company, the better is the product outcome.”

Last but not the least is the user profile the product is intended for. While a lot of companies would hand over the specs to the design house and think that their work is done, some companies go an extra mile and tell the design house who exactly the product is intended for (the target audience). This would not only help the designer in suggesting what could be better, but it will give the contract manufacturing company an exact idea about the quality of materials to put into the product.

“It is relatively easy to marry the design and manufacturing part if the EMS partner understands the user profile the product is intended for. However, if the expectations do not match the intent, the venture becomes unscalable,” explains Balamurugan R, owner, Pulsars.

The requirements and ability

What if the design and manufacturing partner you intend to outsource to is very capable in producing small numbers but is not equipped to do well if large numbers are a part of the game? Or, what if the partner pays